

Fundamentals of Information Systems Security

Chapter 2 Quiz Answer Key

Instructor: Guuleed Maxamuud Aw Abdi

Section A: Multiple Choice (40 points)

1. **C** – Kevin Ashton
2. **B** – RFID
3. **D** – Decreasing internet speed
4. **C** – 70-90%
5. **D** – Cost reduction
6. **C** – Interoperability
7. **C** – Real-time communication
8. **C** – Nanotechnology
9. **B** – They often have limited processing power and security features
10. **A** – Health monitoring wearables

Section B: True/False (20 points)

1. **True** – Information security protects organizational ability to function
2. **True** – 3D printing enables mass customization
3. **False** – Convergence amplifies their individual impact
4. **True** – IoT amplifies existing internet challenges
5. **False** – Law often struggles to keep pace with technological change
6. **True** – B2C stands for Business-to-Consumer
7. **True** – Smart cities use IoT for traffic management
8. **False** – Privacy is a major concern with IoT
9. **True** – RFID to IoT is a significant shift
10. **False** – Telemedicine is enabled by emerging technologies

Section C: Short Answer (40 points)

1. Five Key Drivers for IoT Growth (10 points)

- **Global IP-based Networking:** Universal adoption of Internet protocol provides a ready-made global network
- **Ubiquitous Connectivity:** Widespread availability of broadband Internet
- **Smaller and Faster Computing:** Advancements in semiconductor technology
- **Growth of Cloud Computing:** Scalable storage and computing power
- **Data Analytics:** Ability to capture, analyze, and derive insights from data

2. Five Critical IoT Challenges (10 points)

- (a) **Security:** Protecting devices and data from attackers
- (b) **Privacy:** Who controls data, who owns it, how it is used
- (c) **Interoperability:** Devices from different manufacturers must communicate
- (d) **Legal and Regulatory Compliance:** Liability, data sovereignty
- (e) **Emerging Social and Economic Issues:** Job displacement, digital divide

3. IoT Impact on Human Life and Business (10 points)

- **Human Life:** Health monitoring wearables, home security and smart homes, online calendars and coordination, GPS tracking, online finance and e-commerce
- **Business:** Digital presence (B2C), virtual workplace, remote monitoring and automation, smart city applications, transformation of service delivery

4. Interoperability and Security (10 points)

- Incompatible systems often lead to workarounds that create new security holes
- When devices cannot communicate securely, users may bypass security measures
- Lack of standards can lead to fragmented, insecure ecosystems
- This creates vulnerabilities that attackers can exploit